



5.03 Drilling Rig Orientation

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps for Drilling Rig Orientation .

2. Scope

This procedure is intended for the use by Silverpoint employees .

3. Hazard Identification

- Slips, Trips and Falls
- Personal Injury
- Physical Stress
- Pinch Points
- Equipment Failure and Damage
- Information Overload
- Poor Visibility
- Lifting , Twisting , Pushing , Pulling and Bending
- Backing Up
- Uneven Ground
- Open Pits
- Overhead Lines
- Flare Line

4. EHS Requirements

- F/R Coveralls .
- Steel Toed Boots .



- Safety Glasses .
- Hearing Protection
- Gloves .

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- MED HAT THA 5.03

6. Procedure

- 6.1. When you arrive on a new project site, it is good policy to find the consultant and introduce yourself . Getting to know your consultant is a very good thing since he is the supervisor on that location . Be sure to give him your contact information (phone #) so he can get a hold of you if you are away from the rig. Unless you are working with a Silverpoint tech, the consultant will be signing your field tickets .
 - 6.1.1. Very important to remember that when you arrive on location , stay with-in the surveyed boundary of the lease and access road. This is marked with stakes or colored cones . The oil and gas company has bought this from the landowner and if you go outside the boundary , they may have to pay for the damage caused by the truck to the landowner .
- 6.2. While you are talking to the consultant you should ask him where the best place for you to park your shack would be. If the rig is still setting up, you might have to wait till they are done . Most of the time we put our shacks at the end of the push shack . Some rigs have a generator that we plug into for power, but most of the time we plug into one end of the push shack if you are not sure ask someone to show where to plug in, so you don't damage anything .
- 6.3. After you have unhooked from your shack , drop your hitch by your shack if equipped , so it's ready for when you move . Also, if you don't take it off it will damage your 6 to 4 fitting when you are spreading . Once you have your shack all set up, you can go and unload the hoses you will need for the job. This just shows everyone that you are ready to go to work when called upon . A lot of times when the rig hands come to get you, they are in a hurry for you to go and take a load, so if you have everything ready, it is a lot less stressful on everyone .



- 6.4. Your field tech should be there to show you the spread field, water source and cement pit, if it's not on location . When you are being shown the water source , field or cement pit, it is always preferred that this is done in the daylight , this way you can see if there are any rocks , ruts , holes , cows , bails , old equipment , or anything else that is a potential hazard for when you are working at night . Most of the Silverpoint techs carry flagging or pylons and are more than happy to mark something for you so you don't run into it. If they don't have anything they can get some , the better you know where you are going , the better the field looks when you are done .
- 6.5. Vacuum trucks **DO NOT** load from water sources . If you are asked to haul water , the water truck will have to pump the water into your vacuum truck ; even if you have a brand-new hose , it is considered contaminating the water source .
- 6.6. If you happen to notice any tripping hazards while walking around , report them to the site supervisor , mark them or eliminate them so they don't harm anyone or yourself .
- 6.7. Refer to SOP 1.04 for the names of the rig tanks and their purpose in drilling . This procedure also breaks down how to use some of the equipment that we carry on the truck . It also tells you some of the responsibilities that we have while on the job site . Remember there are areas that you don't need to be on the rig . You are in their work area ; it is always good to ask if you can go somewhere , you are not sure . Stay out of their way so they can do their job without them having to watch out for you .
- 6.8. **Drilling Procedure** : Set surface , drill surface , circulate , run surface casing , circulate surface casing , surface cement , wait on cement , nipple up , drill out surface cement , drill main hole , begin fast hole or flocking , mud up , (depends on depth) , circulate main hole , come out sideways , run main hole casing , circulate casing , main hole cement , tear out .
- 6.9. These are some terms you might hear when working around the rig ; wiper trip , bit trip , coming out sideways , logging , survey , BOP's , nipple up , mouse hole , rat hole .
 - 6.9.1. **Wiper trip** – this is when they pull all the pipe joints that they are drilling with , out of the hole , to clean the hole , on conventional rigs they stand the pipe in the derrick .
 - 6.9.2. **Bit trip** – this is when they pull the entire drilling pipe out of the hole to change the drill bit itself , because it is broken , worn out or just isn't working properly .
 - 6.9.3. **Coming out sideways** – this is simply pulling the pipe out of the hole and laying it down in the pipe racks .
 - 6.9.4. **Logging** – is when they have a wire line truck come in and put a tool down the hole and gives them a reading



of what's down the hole.

- 6.9.5. **Survey** – done every 50 or 100m, to tell them if they are drilling straight. They drop a tool down inside the pipe, attached to a wire, so they can retrieve it.
- 6.9.6. **BOP's** – are attached to the casing after surface is drilled, to have control of the well while drilling main hole. This is a series of valves that can be opened or closed by the driller.
- 6.9.7. **Nipple up** – this is done after they have drilled surface and waited till the cement has set. They put on the BOP's and then start running the bit and drill pipe into the hole.
- 6.9.8. **Mouse hole** – on older rigs, this is a hole that they run their next pipe that is to be attached for drilling into. It usually is under the floor of the rig.
- 6.9.9. **Rat hole** – on older rigs, this runs through the rig floor, down to beside the cellar, into the ground, that they run their top drive into, while they are connecting another piece of drill pipe.

6.10. When you are around the rig, remember that there are areas that you need to be careful of, that can be potentially dangerous. The cellar is an open hole that you could fall into. The rat hole can be a small hole to a big one; this hole can be very deep. When driving around the rig remember that there will be a power line running from the push shack to the rig. There can also be guide wires, flare lines and open pits, as well as equipment, people and other vehicles parked or moving around the lease.

6.11. Rig workers are broken down by job title names, starting from the lowest to the highest job importance, Lease Hand, Roughneck, Motor Hand, Derrick Hand, Driller, and Tool Push.

- 6.11.1. **Lease Hand** – cleans the garbage around the lease, washes the rig and generally helps wherever needed.
- 6.11.2. **Roughneck** – slings most of the pipe on the rig floor, helps where needed and helps the derrick hand when needed.
- 6.11.3. **Motor Hand** – takes care of the generator and is in charge of all the motors on the rig. Makes sure that they have oil changed when needed. Also helps out on the rig floor when needed.
- 6.11.4. **Derrick Hand** – Is in charge of the mud and flock and makes sure that these are the right weight and viscosity. Works directly with the driller and vacuum truck and water truck drivers.
- 6.11.5. **Driller** – this is the rig hand that drills the hole, when working around the rig he is in charge of all the other workers and makes sure that they are doing their job. He works very close with the derrick hand because he needs the mud and flock to be perfect to drill the hole.
- 6.11.6. **Rig Manager /Tool Push** – the tool push is in charge of all the workers on the rig. Normally he knows the rig very well and makes sure that it is running to its best potential. He also answers to the consultant.



Remember : if you are in doubt about what to do, call your Supervisor